Trolley Retrieval: A Complete Analysis with a Machine-Checked Distinguishability Theorem

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Abstract

We analyse the *Trolley Retrieval* problem (G-Research GRiddles Series A, Puzzle 4). A trolley sits on an unknown cell of a one-way infinite tape whose cell n carries the label $c(n)$, where c is a permutation of the positive integers known to the player Bob. Each turn Bob pushes the trolley one cell left or right and is told whether the label of the new cell is strictly less than the current time. Bob must eventually name the trolley's exact position; falling off the left end, or guessing wrongly, loses.

We prove the answer: **Bob has a winning strategy if and only if c is *not* eventually the identity**, i.e. $c(n) \neq n$ for infinitely many n (equivalently, c is not a finitary permutation). We give the full elementary proof of both directions. The losing direction rests on a parity lock: the threshold Bob can probe is always *even*, so an identity tail makes the pair $(2j,\ 2j+1)$ permanently indistinguishable. The winning direction rests on a *distinguishability theorem*: for a non-eventually-identity c, no pair of starting cells is indistinguishable. Its crux—previously believed to require global surjectivity bookkeeping—falls to an elementary forward-closure argument.

The losing direction, the distinguishability theorem, and explicit winning strategies for two natural "locally decodable" families are formalised and machine-checked in Lean 4 and are axiom-clean. We are scrupulous about the boundary of what is proven: a single, uniform, explicitly formalisable winning strategy for *every* non-eventually-identity c remains open, owing to a genuine timing/resolution tension that we describe precisely.

1 The problem

The exact statement is as follows. Let $c(1), c(2), \ldots$ be a sequence of positive integers in which every positive integer appears exactly once; equivalently c is a permutation (bijection) of $\mathbb{N}^+$. A one-way infinite sequence of cells is given: cell n (the n-th from the left, $n \geq 1$) is labelled $c(n)$.

At time $t = 0$ an invisible trolley is placed on an unknown cell, the k_0-th from the left. For each integer $t \geq 1$, in order:

1. Bob chooses a direction, left or right; the trolley moves one cell that way, to cell k_t.
2. The trolley sends Yes if $c(k_t) < t$, and No otherwise.

If ever $k_t = 0$ (the trolley falls off the left end), Bob loses immediately. At any time Bob may instead *guess* the trolley's position; a correct guess retrieves it, an incorrect guess loses.

> *Determine all permutations c for which Bob has a strategy guaranteeing safe retrieval.*

Bob knows c in full; the only hidden datum is the start cell k_0. The trolley is invisible—Bob never sees k_t directly. His sole feedback is the binary signal, and he chooses each move adaptively from the signal history.

1.1 The answer

Theorem 1.1 (Main)**.** *Bob has a strategy ensuring safe retrieval if and only if c is* not eventually the identity*; that is, if and only if $c(n) \neq n$ for infinitely many n.*

We write c is *eventually identity* (EI) if there is a threshold N with $c(n) = n$ for all $n > N$. Since c is a bijection, "not EI" has several equivalent forms, all used below:

- $\{ n : c(n) \neq n \}$ is infinite (c is not finitary);
- c^{-1} is not eventually identity (the condition is symmetric under $c \leftrightarrow c^{-1}$);
- c has infinitely many *upward gaps*: positions p with $c(p{+}1) - c(p) \geq 2$.

The last equivalence is worth recording, as it is exactly what fails in the EI case.

Proposition 1.2. *c has infinitely many upward gaps if and only if c is not eventually identity.*

Proof. If $c(n{+}1) \leq c(n) + 1$ for all $n > N$ (no gaps past N), then on (N, ∞) the bijection c is non-decreasing by steps of at most 1; a strictly value-preserving bijection of a tail of $\mathbb{N}^+$ that never jumps up by 2 cannot skip a value and cannot repeat one, forcing $c(n) = n$ for all $n > N$. The contrapositive is the claim. □

1.2 Reformulation: the alarm-clock view and the parity lock

It is convenient to read the labels as wake-up times. Think of cell n as carrying an alarm clock set for time $c(n)$: at every time $t > c(n)$ the cell is "awake", and a trolley standing on it reports Yes; while $t \leq c(n)$ the cell is "asleep" and reports No. Because c is a bijection, exactly one cell wakes at each integer time, and every cell wakes eventually. This is just a restatement of signal $=$ Yes $\iff c(k_t) < t$.

Let $D_t := k_t - k_0$ be Bob's net displacement at time t.

Lemma 1.3 (Parity lock)**.** *As long as Bob has not yet guessed, $D_t \equiv t \pmod 2$. Equivalently, $t - D_t$ is always even.*

Proof. $D_0 = 0$, and each move changes D by ± 1 and t by $+1$, so the parity of $t - D_t$ is invariant; it starts even. □

The parity lock is the single structural fact behind the losing direction. A second, equally elementary identity exposes *why*. Let L_t be the number of *left* moves Bob has made by time t and R_t the number of right moves, so $t = L_t + R_t$ and $D_t = R_t - L_t$. Then

$$t - D_t \; = \; (L_t + R_t) - (R_t - L_t) \; = \; 2L_t. \tag{1}$$

Write $g(m) := c(m) - m$ for the displacement of the label from the diagonal. The signal at time t is

$$\begin{aligned} \text{Yes} \iff c(k_0 + D_t) < t &\iff g(k_0 + D_t) < t - D_t - k_0 \\ &\iff g(k_0 + D_t) < 2L_t - k_0. \end{aligned} \tag{2}$$

In other words, the walk reads the sequence g against a threshold $2L_t - k_0$ that Bob controls only through his *left-move count*.

Remark 1.4 (The whole obstruction, in one line)**.** *For the identity, $g \equiv 0$, so (2) collapses to Yes $\iff k_0 < 2L_t$. Bob can only ever compare k_0 against* even *thresholds, and $2j < 2L \iff 2j{+}1 < 2L$ for every L. Hence the start cells $2j$ and $2j{+}1$ produce* identical signals under every strategy*: the parity bit of k_0 is unrecoverable. This is the seed of the losing direction; the work below is to extend it to every eventually-identity c despite the perturbed initial segment.*

2 Losing direction: eventually identity ⇒ Bob loses

Theorem 2.1. *If c is eventually identity, Bob has no strategy that wins from every start cell.*

The proof isolates a structural "parallel-evolution" lemma and then verifies its hypothesis for two carefully chosen start cells.

2.1 Parallel evolution

Fix a strategy σ. For a start cell k let $\mathrm{pos}(k,t)$ be the position and $\mathrm{hist}(k,t)$ the signal history (most-recent first) at time t. Bob's action at any decision point is a function of hist alone.

Lemma 2.2 (Parallel evolution)**.** *Let k_1, k_2 be two start cells. Suppose that, started from k_1, Bob makes no guess before time t, and that at every time $s \le t$ the signal observed from k_1's trajectory equals the signal observed from k_2's trajectory. Then the two histories agree at time t, and the positions stay rigidly offset:*

$$\mathrm{hist}(k_1,t) = \mathrm{hist}(k_2,t), \qquad \mathrm{pos}(k_2,t) = \mathrm{pos}(k_1,t) + (k_2 - k_1).$$

Proof. Induction on t. At $t = 0$ the histories are empty and the offset is $k_2 - k_1$ by definition. For the step, equal histories make σ choose the *same* action in both worlds; since Bob has not guessed, that action is a move, applied identically to both, preserving the offset. The new signal entries are equal by the time-$(t{+}1)$ hypothesis, so the histories remain equal. □

Corollary 2.3. *If $k_1 \ne k_2$ and the two signal traces agree at* every *time under σ, then σ cannot win from both k_1 and k_2.*

Proof. Equal traces force equal histories at all times (Lemma 2.2 applied up to either guess time), hence equal first-guess times $T_1 = T_2 =: T$ and equal guesses $g_1 = g_2$. But correctness would require $g_1 = \mathrm{pos}(k_1,T)$ and $g_2 = \mathrm{pos}(k_2,T)$, while $\mathrm{pos}(k_2,T) = \mathrm{pos}(k_1,T) + (k_2 - k_1)$. These contradict $k_1 \ne k_2$. □

2.2 The indistinguishable pair for eventually-identity c

Suppose $c(n) = n$ for all $n > N$. Let M be an upper bound for the finitely many perturbed labels: $M \ge \max\{\, c(m) : 1 \le m \le N \,\}$ and $M \ge N + 1$. Choose j with $2j > M$ and set

$$k_1 = 2j, \qquad k_2 = 2j + 1.$$

Lemma 2.4 (Signal match)**.** *Fix a no-guess strategy run from $k_1 = 2j$. At every time t, every position $p \ge 1$ reachable from $2j$ in t moves (so $2j - t \le p$) that respects the parity lock ($t + p$ even) satisfies*

$$\textit{signal at } p \textit{ at time } t \;=\; \textit{signal at } p{+}1 \textit{ at time } t.$$

Proof. Two cases on the location p.

(a) $p > N$ (identity region). Both p and $p{+}1$ exceed N, so $c(p) = p$ and $c(p{+}1) = p{+}1$. The signal at p is Yes $\iff p < t$, and at $p{+}1$ is Yes $\iff p{+}1 < t$. These differ only if $t = p + 1$. But then $t + p = 2p + 1$ is odd, violating the parity hypothesis $t + p$ even. So the signals agree.

(b) $p \le N$ (perturbed region). Here the trolley has travelled from $2j$ to $p \le N$, so $t \ge 2j - p \ge 2j - N > M - N \ge \dots$; concretely $2j > M \ge N + 1$ forces $t > M$. Both $c(p)$ and $c(p{+}1)$ are at most M (using $c(N{+}1) = N{+}1 \le M$ for the boundary), hence $c(p), c(p{+}1) < t$: both signals are Yes. They agree. □

The two cases capture the dichotomy of Remark 1.4: in the identity tail, the parity lock makes $2j$ and $2j{+}1$ collide; in the perturbed segment, the time cost of reaching it ($t \ge 2j - N$) is so large that every cell there is already awake, so both worlds report Yes and again collide.

Proof of Theorem 2.1. Suppose σ wins from every start cell, in particular from $k_1 = 2j$ and $k_2 = 2j{+}1$ above. We claim the two signal traces agree at every time, which contradicts Corollary 2.3.

By induction on t, the trajectories from $2j$ and $2j{+}1$ evolve in lockstep with positions offset by exactly 1 and equal histories, as long as Bob has not guessed: the inductive step applies the signal match (Lemma 2.4) at $\mathrm{pos}(2j, t{+}1)$, whose hypotheses—validity $p \geq 1$ (Bob wins, so the trajectory is safe), reachability $2j - t \leq p$ (a ± 1 walk), and parity (Lemma 1.3)—all hold. Hence the signal entering both histories at each step is identical, so $\mathrm{hist}(2j, t) = \mathrm{hist}(2j{+}1, t)$ and $\mathrm{pos}(2j{+}1, t) = \mathrm{pos}(2j, t) + 1$ for all t. The traces agree at every time, and Corollary 2.3 gives the contradiction. □

Remark 2.5 (The half-infinite tape is irrelevant here)**.** *For the chosen large $k_0 \approx 2j$, no finite-time strategy can reach cell 0, so the "falling off" rule never binds; the indistinguishability argument is unaffected by the absence of a left end.*

3 Winning direction: not eventually identity ⇒ Bob wins

The heart of the winning direction is a converse to the losing mechanism: when c is not eventually identity, *no* pair of start cells can stay indistinguishable. We make "indistinguishable" precise, prove the theorem, and then explain how distinguishability is turned into a winning strategy—fully for two natural families, and with an honest account of the general gap.

3.1 The distinguishability theorem

Reaching displacement D from the start costs time at least $|D|$, and the parity lock forces $t \equiv D \pmod 2$. Within those constraints Bob can pad with oscillation to reach any admissible time, and (because a cell once awake stays awake) the only information at displacement D is the *set of times $t \geq D$, $t \equiv D \pmod 2$, at which the cell is awake*—equivalently, the single comparison $c(a + D) < t$. This motivates:

Definition 3.1 (Indistinguishable pair)**.** *Two start cells $a < b$ are* indistinguishable *(for c) if for every displacement $D \geq 0$ and every time $t \geq D$ with $t \equiv D \pmod 2$,*

$$c(a + D) < t \iff c(b + D) < t.$$

Quantifying over *all* reachable (D, t)—including small times that catch early wake-ups—is essential; it is exactly the freedom that lets Bob's oscillation observe a cell's wake-up time at full resolution.

Theorem 3.2 (Distinguishability)**.** *If $a < b$ are indistinguishable for c, then $b = a + 1$ and $c(n) = n$ for all $n \geq a$. Consequently, if c is not eventually identity, then* no *pair of start cells is indistinguishable.*

We prove the first sentence; the second is its contrapositive (an indistinguishable pair would exhibit an identity tail, making c EI). Throughout write $\Delta := b - a \geq 1$.

The proof has two regimes by the size of Δ. The case $\Delta = 1$ is a clean shift argument; the case $\Delta \geq 2$ is the crux. We first extract the local structure shared by both.

Lemma 3.3 (Consecutiveness)**.** *Call a displacement D* active *(for the cell a) if $c(a + D) \geq D$, and similarly for b. If $a < b$ are indistinguishable and D is active for both a and b, then $c(a + D)$ and $c(b + D)$ are consecutive integers:*

$$|\, c(a + D) - c(b + D) \,| = 1.$$

Proof. Let $X = c(a + D)$, $Y = c(b + D)$. Injectivity gives $X \neq Y$; say $X < Y$. If $Y \geq X + 2$, pick t with $X < t \leq Y$ of the parity $t \equiv D \pmod 2$ (possible because the interval $(X, Y]$ has length ≥ 2 and so contains both parities). Then $c(a + D) < t$ but $c(b + D) \not< t$, with $t \geq D$ (as D is active and $t > X \geq D$), contradicting indistinguishability at (D, t). Hence $Y = X + 1$. The case $Y < X$ is symmetric. □

Lemma 3.4 (Bad displacements are forward-closed)**.** *Call D* bad *(for a) if $c(a + D) < D$. If $a < b$ are indistinguishable and D is bad, then $D + \Delta$ is bad: $c(a + D + \Delta) < D + \Delta$.*

Proof. D bad means $c(a + D) < D$, i.e. the cell $a + D$ is awake at the admissible time t with $t \equiv D \pmod 2$, $D \leq t$, as soon as $t > c(a{+}D)$; take the least such admissible t, so $t \leq D{+}1$ and the signal is Yes. Indistinguishability transfers Yes to $b + D = a + (D + \Delta)$: $c(a + D + \Delta) < t \leq D + 1$, and since the value is an integer $c(a + D + \Delta) \leq D < D + \Delta$. (The point is purely that an already-saturated Yes propagates by $+\Delta$; this is where the constraint $b = a + \Delta$ enters.) □

Lemma 3.4 is decisive: on each residue class modulo Δ, the bad displacements form an *up-set*, so the *active* (non-bad) displacements form a finite or cofinite *prefix*—never an interleaving of good and bad. This is exactly the structural fact that earlier, more elaborate attempts (parity arguments, value-hogging bookkeeping) were trying to recover. With it, the crux is short.

Lemma 3.5 (Chain bound)**.** *If $a < b$ are indistinguishable, fix a residue r with $0 \leq r < \Delta$. For any k such that $r, r + \Delta, \ldots, r + k\Delta$ are all active,*

$$c(a + r + k\Delta) \ \leq\ c(a + r) + k.$$

Proof. By Lemma 3.3, consecutive active displacements on the residue satisfy $|c(a+r+(i+1)\Delta) - c(a + r + i\Delta)| = 1$ (apply consecutiveness at the displacement $r + i\Delta$, whose $+\Delta$ shift lands at $b+r+i\Delta$). Summing k unit steps and the triangle inequality give $c(a+r+k\Delta) \leq c(a+r)+k$. □

Lemma 3.6 ($\Delta \geq 2$ is impossible)**.** *If $a < b$ are indistinguishable with $\Delta = b - a \geq 2$, contradiction.*

Proof. Let $M := \max_{0 \leq r < \Delta} c(a + r)$. Suppose a displacement D is active, and write $D = r + k\Delta$ with $0 \leq r < \Delta$. By forward-closure (Lemma 3.4), *all* of $r, r + \Delta, \ldots, r + k\Delta$ are active (active is a prefix on each residue). Activity gives $c(a + D) \geq D = r + k\Delta$, while the chain bound gives $c(a + D) \leq c(a + r) + k \leq M + k$. Hence

$$r + k\Delta \leq M + k \quad \Longrightarrow \quad k(\Delta - 1) \leq M - r \leq M.$$

With $\Delta \geq 2$ this bounds $k \leq M$, so $D = r + k\Delta < (M+1)\Delta$. Thus *every* displacement $D \geq (M+1)\Delta$ is bad: $c(a + D) < D$ for all such D.

This forces a pigeonhole collapse. For $D \geq (M+1)\Delta$, the cell at position $a + D$ has label $c(a + D) < D = (a + D) - a$, i.e. $c(p) < p - a$ for all large p. The finitely many small positions have labels bounded by some constant. Choosing N large enough, c maps the segment $[1, N]$ into $[1, N - a - 1]$: large positions p land below $p - a \leq N - a - 1$, and the bounded small positions also fit once N is large. But $[1, N]$ has N elements and $[1, N - a - 1]$ has $N - a - 1 < N$, so c cannot be injective on $[1, N]$. Contradiction. □

Remark 3.7 (Why this is the right proof)**.** *The contradiction in Lemma 3.6 uses* no *parity argument and* no *even/odd split on Δ. The single lever is forward-closure of badness (Lemma 3.4), which turns the "active" set on each residue into a prefix; the chain bound then makes that prefix finite for $\Delta \geq 2$, and a one-line pigeonhole finishes. This is markedly simpler than—and subsumes—the earlier even/odd case analysis.*

It remains to handle $\Delta = 1$.

Lemma 3.8 ($\Delta = 1$ forces an identity tail)**.** *If $a < b = a + 1$ are indistinguishable, then $c(n) = n$ for all $n \geq a$.*

Proof. For $\Delta = 1$, $b + D = a + (D + 1)$, so consecutiveness (Lemma 3.3) on active displacements reads $|c(a + D + 1) - c(a + D)| = 1$: the labels along $a, a{+}1, a{+}2, \ldots$ form a self-avoiding ± 1 walk. Let $\sigma(0) = \operatorname{sgn}\big(c(a{+}1) - c(a)\big)$ be the sign of the first step. If $\sigma(0) = -1$ (descending start), the walk $c(a), c(a){-}1, c(a){-}2, \ldots$ would eventually drop below 1, impossible for labels in $\mathbb{N}^+$; so $\sigma(0) = +1$. A self-avoiding ± 1 walk that starts ascending must ascend forever (any later descent would revisit a value, breaking injectivity), giving $c(n + 1) = c(n) + 1$ for all $n \geq a$, i.e. $c(n) = n + C$ on $[a, \infty)$ with $C = c(a) - a$. If $C > 0$, the values $\{1, \ldots, C\}$ are absent from $c([a, \infty))$ and must be supplied by the finite set $c([1, a{-}1])$, which has only $a{-}1$ elements; bijectivity of c on $\mathbb{N}^+$ then forces $C = 0$ (and $C < 0$ was already excluded). Hence $c(n) = n$ for $n \geq a$. (In Lean this sign analysis is the pair `indist_Delta1_pos` / `indist_Delta1_neg`.) □

Proof of Theorem 3.2. Let $a < b$ be indistinguishable, $\Delta = b - a$. By Lemma 3.6, $\Delta \geq 2$ is impossible, so $\Delta = 1$, i.e. $b = a + 1$; then Lemma 3.8 gives $c(n) = n$ for $n \geq a$, so c is eventually identity. Contrapositively, if c is not eventually identity then no pair is indistinguishable. □

3.2 From distinguishability to a strategy

Theorem 3.2 says that for a non-EI c every pair of cells admits a separating observation. A winning strategy must (i) narrow the start cell down to a small candidate set, then (ii) reach a separating observation while keeping the trolley on the tape. Safety is free if Bob only ever oscillates and moves right: positions stay $\geq k_0 \geq 1$, so the trolley never falls off.

Phase 1 (bracket to two candidates). Bob oscillates the displacement between 0 and 1, moving *right first*, so $D_t \in \{0,1\}$ and $k_t \geq k_0$. At every even time $t = 2m$ the trolley sits on k_0, with signal $\textsf{Yes} \iff c(k_0) < 2m$. Since $c(k_0)$ is finite, there is a first even time $T_1 = 2m^*$ with a $\textsf{Yes}$, pinning $c(k_0) \in \{T_1{-}2,\ T_1{-}1\}$, hence

$$k_0 \in \{\, c^{-1}(T_1{-}2),\ c^{-1}(T_1{-}1)\,\}$$

(a singleton if one of those preimages is 0, in which case Bob guesses at once). So Phase 1 always narrows k_0 to (at most) two candidates, the cells waking at one even and the next odd time.

Phase 2 (resolve). Bob must now separate the two candidates. By Theorem 3.2 the candidate pair is distinguishable: some admissible (D,t) with $t \geq D$, $t \equiv D \pmod 2$ yields different signals in the two worlds. Reaching it and reading the signal identifies k_0 and hence the current position, which Bob guesses.

The subtlety, and the location of the only remaining gap, is in Phase 2: *which* separating observation to aim for, and how to monitor it at the necessary time resolution. We make this precise in Section 4 after first giving complete strategies for two natural families where Phase 2 is finite and local.

3.3 Two complete winning families (locally decodable)

When the candidate pair is always separated by a wake-up that occurs at a *bounded* displacement, Bob can resolve it by a fixed-amplitude oscillation that monitors a small window from time 0—catching even the early wake-ups that a navigate-then-observe approach would arrive too late to see. Two regimes are formalised.

Definition 3.9 (Locally $K{=}2$ decodable)**.** *c is* locally decodable *if the map $k_0 \mapsto \big($first even $\textsf{Yes}$ time of $c(k_0)$, first odd $\textsf{Yes}$ time of $c(k_0{+}1)\big)$ is injective; equivalently the bucketed pair $(c(k_0), c(k_0{+}1))$ determines k_0.*

Theorem 3.10 ($K{=}2$ winning theorem)**.** *If c is locally decodable, Bob wins. The strategy oscillates $D \in \{0,1\}$, reads off $c(k_0)$ and $c(k_0{+}1)$ from the first even and first odd $\textsf{Yes}$, decodes k_0, and guesses its current position—never leaving $\{k_0, k_0{+}1\}$, so safety is automatic.*

Example 3.11 (Adjacent transpositions)**.** *Let c swap $2i{-}1 \leftrightarrow 2i$ for every i (so $c(n) = n + 1$ if n is odd, $c(n) = n - 1$ if n is even). Here $g(n) = c(n) - n \in \{+1, -1\}$ is bounded, yet c is not eventually identity; Bob wins by local decoding. This already shows the boundary is non-EI, not unbounded growth of g.*

Example 3.12 (Block 3-cycles on consecutive triples)**.** *Let c cyclically rotate each triple $\{3i{-}2, 3i{-}1, 3i\}$, e.g. $3i{-}2 \mapsto 3i{-}1 \mapsto 3i \mapsto 3i{-}2$. This is locally decodable (the bucketed pair of adjacent labels recovers k_0), and Bob wins.*

Some non-EI permutations need a third probed label. The block 3-cycle written "forwards" is the canonical example.

Definition 3.13 (Locally $K{=}3$ decodable)**.** *c is* locally 3-decodable *if the triple of first-$\textsf{Yes}$ times at displacements $0, 1, 2$—equivalently the bucketed triple $(c(k_0), c(k_0{+}1), c(k_0{+}2))$—determines k_0.*

Theorem 3.14 ($K{=}3$ winning theorem)**.** *If c is locally 3-decodable, Bob wins, via a period-4 triangle-wave sweep of displacements $0, 1, 2, 1, 0, 1, 2, \ldots$ with three detectors (first $\textsf{Yes}$ at displacements 0, 1, 2). The displacement stays in $[0, 2]$, so safety is automatic.*

Example 3.15 (Forward block 3-cycle)**.** *Let $c(m) = m{+}1$ unless $3 \mid m$, in which case $c(m) = m{-}2$ (so each block $\{3b{+}1, 3b{+}2, 3b{+}3\}$ rotates forwards: $c(1) = 2$, $c(2) = 3$, $c(3) = 1$). Then $c(m) \leq m + 1$ always, so* no *rightward probe $[c(k_0{+}D) < T_1{+}D]$ ever separates the Phase-1 candidates—a navigate-only strategy fails. But cell 3 wakes early, at time 2 (since $c(3) = 1$), which splits $k_0 = 1$ from $k_0 = 2$ at displacement 1: at ($D{=}1$, $t{=}3$), the world $k_0 = 2$ stands on the* *already-awake cell* 3 *and reports* Yes*, while* $k_0 = 1$ *stands on cell* 2*, still asleep* ($c(2) = 3 \not< 3$)*, and reports* No*. The* $K{=}3$ *sweep, monitoring small displacements from time* 0*, catches this early wake-up. So* c *is locally* 3*-decodable, and Bob wins.*

Example 3.15 is instructive: it is a concrete non-EI Bob-win whose resolution *requires* catching an early wake-up at a small displacement, which is precisely the regime the local decoders are built for—and precisely what naive "oscillate then navigate right" strategies miss.

3.4 Worked examples table

Permutation c	**EI?**	**Bob**	**Reason**
Identity $c(n) = n$	yes	**loses**	$(2j,\, 2j{+}1)$ collide: only even thresholds probeable (Rem. 1.4).
Finitary (any finite-support perm.)	yes	**loses**	EI with $N =$ support bound; same collision for large k_0.
Adjacent transpositions	no	**wins**	locally 2-decodable (Ex. 3.11); bounded g.
Block 3-cycle (rotated triples)	no	**wins**	locally 2- or 3-decodable (Ex. 3.12, 3.15).
Any c rearranging arbitrarily large values	no	**wins**	not EI; distinguishable pairs (Thm. 3.2).

Table 1: Outcomes by family. Eventually-identity $\iff$ Bob loses.

4 Machine-verification status

The argument has been formalised in Lean 4 (with Mathlib) in the development `griddles-p4-lean`. We report the status honestly: a substantial core is machine-checked and axiom-clean, while a single component—an explicit uniform winning strategy for the fully general case—remains open. We do not overclaim.

For each item below, "axiom-clean" means the Lean `#print axioms` audit lists only the three standard foundational axioms `propext`, `Classical.choice`, `Quot.sound`, with *no* `sorryAx` and no custom axioms. The entire library builds (`lake build` succeeds); the only `sorry` in the whole development is the one noted as open below.

What is proven. The losing direction is complete: eventually-identity c admits the indistinguishable pair $(2j, 2j{+}1)$, machine-verified through the parallel-evolution lemma and the signal-match case analysis. The distinguishability theorem is complete and axiom-clean, including its crux for $\Delta \geq 2$: the elementary forward-closure argument of Section 3 (which earlier, by our own assessment, was believed to need global surjectivity bookkeeping). The two locally-decodable winning classes are complete, with the three concrete instances `adjPerm`, `cyc3Perm`, `blkPerm` each proven a Bob-win.

What is open—and why it is genuinely hard. The remaining gap is a single, explicit, uniformly-described strategy that turns "no pair is indistinguishable" (Thm. 3.2) into a formalised `BobWins` for *every* non-EI c. The difficulty is a real *timing/resolution/reach* tension, not a missing lemma:

- Bob occupies one displacement per tick, so he can finely monitor (period ≈ 2) only $O(1)$ displacements at once. Monitoring displacements $0, \ldots, K$ simultaneously by a triangle wave costs time resolution $\approx 2K$, which can blur a discriminator whose separating window is $O(1)$.
- A discriminator that is an *early* wake-up at displacement D must be monitored finely *from time* 0 (it cannot be reached later: Example 3.15).
- A discriminator that is a *far* wake-up requires *reaching* a large displacement, which precludes finely monitoring all small displacements meanwhile.

Component	Status	Lean location
Losing direction (Thm. 2.1)	**proven, axiom-clean**	`Losing.EventuallyIdentity_loses`; parity lock in `Parity.lean`.
Distinguishability (Thm. 3.2)	**proven, axiom-clean**	`LemmaA.Indistinguishable_implies_eventually_identity`; crux `indist_Delta_ge_2_impossible`.
K=2 winning class (Thm. 3.10)	**proven, axiom-clean**	`WinningRestricted.LocallyDecodable_BobWins`; instances `adjPerm_BobWins`, `cyc3Perm_BobWins`.
K=3 winning class (Thm. 3.14)	**proven, axiom-clean**	`WinningKProbe.LocallyK3Decodable_BobWins`; instance `blkPerm_BobWins`.
Capstone (locally-decodable consistency)	**proven, axiom-clean**	`Answer.locallyDecodable_main`.
Main *iff*, winning half, general c	**open**	`Answer.main` ($\Leftarrow$ direction is a `sorry`).

Table 2: Formalisation status. Everything except the general winning strategy is machine-checked and axiom-clean.

A single uniform strategy must reconcile these. We verified by direct simulation that several natural candidate strategies are each incomplete on a concrete family: a navigate-only "monotone-right" strategy fails on the forward block 3-cycle (Example 3.15); "K=2 or navigate" misses mid-range early discriminators of a block 5-cycle; a sequential staircase that probes displacement 0 first arrives at displacement 1 too late to catch an early wake-up. The existence of *some* winning strategy for every non-EI c is confirmed by an exhaustive belief-state (AND/OR) game solver with real safety, which reports LOSE for eventually-identity inputs and WIN from full ignorance for non-EI inputs, stably as the search horizon grows. But a single explicit, provably-complete, formalisable construction is an intricate open problem—plausibly the intended hard core of the puzzle.

Summary. The *answer* (Theorem 1.1) is settled, with the losing direction and the distinguishability theorem machine-checked and axiom-clean, and explicit machine-checked winning strategies for natural families. The one component not yet formalised is a uniform winning strategy for arbitrary non-EI c; its existence is established by an exhaustive game solver, while an explicit formalisable strategy remains open.

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